Algebra_Prac_Groups_Modulo

Zadanie	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Punkty															
Zadanie	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**			**Suma**
Punkty															

Zadanie 1:

Exercise 4. For each expression, describe all integer solutions and any conditions on the parameters. Assume n ≥ 2:

a) $x + k \equiv 0 \pmod{n}$

Zadanie 2:

Exercise 4. For each expression, describe all integer solutions and any conditions on the parameters. Assume n ≥ 2:

b) $cx \equiv 0 \pmod{n}$, where $\gcd(c, n) = 1$

Zadanie 3:

Exercise 4. For each expression, describe all integer solutions and any conditions on the parameters. Assume n ≥ 2:

c) $ax \equiv b \pmod{n}$, where $\gcd(a, n) = 1$

Zadanie 4:

Exercise 4. For each expression, describe all integer solutions and any conditions on the parameters. Assume n ≥ 2:

d) $ax \equiv b \pmod{n}$, where $\gcd(a, n) = d$

Zadanie 5:

Exercise 4. For each expression, describe all integer solutions and any conditions on the parameters. Assume n ≥ 2:

e) $x \equiv a \pmod{n}$ and $x \equiv b \pmod{n}$

Zadanie 6:

Exercise 18. Prove each statement for an arbitrary group (G, □) with identity e:

a) The identity element e is unique

Zadanie 7:

Exercise 18. Prove each statement for an arbitrary group (G, □) with identity e:

b) Every element has a unique inverse

Zadanie 8:

Exercise 18. Prove each statement for an arbitrary group (G, □) with identity e:

c) If a □ b = a □ c, then b = c

Zadanie 9:

Exercise 18. Prove each statement for an arbitrary group (G, □) with identity e:

d) The inverse of the inverse of a equals a

Zadanie 10:

Exercise 18. Prove each statement for an arbitrary group (G, □) with identity e:

e) The inverse of (a □ b) is b^{-1} □ a^{-1}

Zadanie 11:

Exercise 11. On $\mathbb{Z}_n$, let ⊕ and ⊙ denote addition and multiplication modulo n, where n ≥ 2. Prove:

a) 0 ⊕ a = a for every a ∈ $\mathbb{Z}_n$

Zadanie 12:

Exercise 11. On $\mathbb{Z}_n$, let $\oplus$ and $\odot$ denote addition and multiplication modulo n, where n ≥ 2. Prove:

b) $1 \odot a = a$ for every $a \in \mathbb{Z}_n$

Zadanie 13:

Exercise 11. On $\mathbb{Z}_n$, let $\oplus$ and $\odot$ denote addition and multiplication modulo n, where n ≥ 2. Prove:

c) For every $a \in \mathbb{Z}_n$, there is a $b \in \mathbb{Z}_n$ with $a \oplus b = 0$

Zadanie 14:

Exercise 11. On $\mathbb{Z}_n$, let $\oplus$ and $\odot$ denote addition and multiplication modulo n, where n ≥ 2. Prove:

d) If $a \oplus b = 0$, express b in terms of a and n

Zadanie 15:

Exercise 11. On $\mathbb{Z}_n$, let $\oplus$ and $\odot$ denote addition and multiplication modulo n, where n ≥ 2. Prove:

e) The additive identity and additive inverse are unique

Zadanie 16:

Exercise 19. Solve these short problems using modular arithmetic:

a) A clock shows 09:00. What hour will it show 100 hours later?

Zadanie 17:

Exercise 19. Solve these short problems using modular arithmetic:

b) Today is a Tuesday. What weekday will it be 250 days later?

Zadanie 18:

Exercise 19. Solve these short problems using modular arithmetic:

c) A code is the final digit of 7^{34}. Determine the code.

Zadanie 19:

Exercise 19. Solve these short problems using modular arithmetic:

d) An integer leaves remainder 4 upon division by 9. Find the remainder of 5 times it plus 3.

Zadanie 20:

Exercise 19. Solve these short problems using modular arithmetic:

e) Can an integer square have remainder 3 when divided by 4? Prove your answer.

Zadanie 21:

Exercise 14. Compute the least nonnegative residue. Explain how you simplify the power:

a) $2^{20} \bmod 7$

Zadanie 22:

Exercise 14. Compute the least nonnegative residue. Explain how you simplify the power:

b) $3^{15} \bmod 11$

Zadanie 23:

Exercise 14. Compute the least nonnegative residue. Explain how you simplify the power:

c) $5^{12} \bmod 13$

Zadanie 24:

Exercise 14. Compute the least nonnegative residue. Explain how you simplify the power:

d) $7^{8} \bmod 10$

Zadanie 25:

Exercise 14. Compute the least nonnegative residue. Explain how you simplify the power:

e) $4^{25} \bmod 9$

Zadanie 26:

Exercise 14. Compute the least nonnegative residue. Explain how you simplify the power:

f) $6^{100} \bmod 11$